Lecture 9: Quadratic Optimization
KKT Optimality Conditions

Oct 20, 2026

Quick Announcements

- My office hours: always on Thursdays 4-5:30pm (same Google cal link)
- Thursday (Oct 29) - midterm exam
- Agenda for today
  - Duality in Quadratic Optimization
  - A tiny bit of Saddle Theory
  - KKT Optimality Conditions
  - Fenchel duality

Last Time: Convex Duality Framework

$$\begin{aligned} \text{minimize}_{x \in X} \ & f_0(x) \\ \text{subject to } & f_i(x) \leq 0, \quad i = 1, \ldots, m, \\ & h_j(x) = 0, \quad j = 1, \ldots, s \\ \text{variable } & x \in \mathbb{R}^n \end{aligned}$$

- With λ_i, ν_j denoting Lagrange multipliers for g_i, h_j, respectively, Lagrangian is:

$$\mathcal{L}(x, \lambda, \nu) = f_0(x) + \sum_{i=1}^{m} \lambda_i f_i(x) + \sum_{j=1}^{s} \nu_j h_j(x),$$

- With $g(\lambda, \nu) := \inf_{x \in X} \mathcal{L}(x, \lambda, \nu)$, the dual problem becomes:

$$\begin{aligned} &\text{maximize } g(\lambda, \nu) \\ &\text{subject to } \lambda \geq 0. \end{aligned}$$

- For a **convex optimization problem** (f_0, f_i convex, h_j affine), strong duality holds if the **Slater condition** holds: $\exists x \in \operatorname{rel\,int}(X)$ such that $f_i(x) < 0$ for $i = 1, \ldots, m$

QPs and QCQPs

Quadratic Programs

A **Quadratic Program (QP)** is an optimization problem of the form:

$$\begin{aligned} \min \ & \frac{1}{2} x^T Q x + c^T x \\ & A_1 x = b_1 \\ & A_2 x \leq b_2 \end{aligned}$$

where $Q = Q^T$.

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Quadratically Constrained Quadratic Programs

A **Quadratically Constrainted Quadratic Program (QCQP)** is a problem:

$$\begin{aligned} \min \ & \frac{1}{2} x^T Q_0 x + c^T x \\ & x^T Q_i x + q_i^T x + b_i \leq 0, i = 1, \ldots, m \\ & Ax = b \end{aligned}$$

where $Q_i, i = 0, \ldots, m$ are **symmetric** matrices.

Convex if $Q_0 \succeq 0, Q_i \succeq 0$. Gurobi can now handle **non-convex** QCQPs!

One Problem to Warm Up

Convex QCQP

$$\begin{aligned} \text{minimize } & \frac{1}{2}x^T Q_0 x + q_0^T x + r_0 \\ \text{subject to } & \frac{1}{2}x^T Q_i x + q_i^T x + r_i \leq 0, \quad i = 1, \ldots, m, \end{aligned}$$

where $Q_0 \succ 0$ and $Q_i \succeq 0$

- **What is the Lagrangian? What is the dual? Does Slater Condition hold?**

Quadratic Programs - Preliminaries

Unconstrained Quadratic Program

For $Q = Q^{\mathsf{T}}$, consider the following unconstrained problem:

$$\min\ f(x) := \frac{1}{2} x^{\mathsf{T}} Q x + q^{\mathsf{T}} x$$

- **What is the optimal value $p^\star$?**

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- **What is the optimal value $p^\star$?**

$$\nabla_x f(x) = 0 \quad \Leftrightarrow \quad Qx = -q$$

$$p^\star = \begin{cases} & \text{if } Q \succeq 0 \text{ and } q \in \mathcal{R}(Q) \\ -\infty & \text{otherwise.} \end{cases}$$

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$$x^\star = -Q^\dagger q + z \text{ with } z \in \mathrm{Null}(Q)$$

$$p^\star = \begin{cases} & \text{if } Q \succeq 0 \text{ and } q \in \mathcal{R}(Q) \\ -\infty & \text{otherwise.} \end{cases}$$

- $Q^\dagger$: Moore-Penrose pseudo-inverse

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  - for Q with singular value decomposition $Q = U\Sigma V^{\mathsf{T}}$, $Q^\dagger := V\Sigma^\dagger U^{\mathsf{T}}$
  - $\Sigma^\dagger$ reciprocates all non-zero singular values of Σ and leaves zeros as zeros

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Convex QCQP

QCQP

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where $Q_0 \succ 0$ and $Q_i \succeq 0$

- The Lagrangian is:

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where $Q_0 \succ 0$ and $Q_i \succeq 0$

- The Lagrangian is:

$$\mathcal{L}(x, \lambda) = \frac{1}{2}x^T Q(\lambda) x + q(\lambda)^T x + r(\lambda),$$

where $Q(\lambda) = Q_0 + \sum_{i=1}^m \lambda_i Q_i, \quad q(\lambda) = q_0 + \sum_{i=1}^m \lambda_i q_i, \quad r(\lambda) = r_0 + \sum_{i=1}^m \lambda_i r_i$

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- Because $\lambda \geq 0$, we have $Q(\lambda) \succ 0$ and therefore:
$$g(\lambda) = \inf_x L(x, \lambda) = -\frac{1}{2}q(\lambda)^T Q(\lambda)^{-1} q(\lambda) + r(\lambda).$$
- We can express the dual problem as:
$$\max_{\lambda \geq 0} -\frac{1}{2}q(\lambda)^T Q(\lambda)^{-1} q(\lambda) + r(\lambda)$$

Convex QCQP

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$$\mathcal{L}(x, \lambda) = \frac{1}{2}x^T Q(\lambda) x + q(\lambda)^T x + r(\lambda),$$
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- We can express the dual problem as:
$$\max_{\lambda \geq 0} -\frac{1}{2}q(\lambda)^T Q(\lambda)^{-1} q(\lambda) + r(\lambda)$$
- Slater condition holds if there exists an x with
$$\frac{1}{2}x^T Q_i x + q_i^T x + r_i < 0, \quad i = 1, \ldots, m.$$

Other Important Examples in the Notes

- A **non-convex** QCQP: for $Q = Q^{\mathsf{T}}$ and $Q \not\succeq 0$, consider:

$$\begin{aligned} &\text{minimize } x^{\mathsf{T}} Q x + 2c^{\mathsf{T}} x \\ &\text{subject to } x^{\mathsf{T}} x \leq 1 \end{aligned}$$

  This has a nice "hidden convexity" property

- Regularized Support Vector Machines (SVM)

- Entropy Maximization

Saddle Point Theory

- Optional reading in the notes, but very insightful

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Alternative Formulation of Primal and Dual Problems

We can express **the optimal values of the primal and dual** as:

$$p^\star = \inf_{x \in X} \sup_{\lambda \geq 0} \mathcal{L}(x, \lambda) \qquad d^\star = \sup_{\lambda \geq 0} \inf_{x \in X} \mathcal{L}(x, \lambda)$$

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- **Weak duality** restatement:

$$\sup_{\lambda \geq 0} \inf_{x \in X} \mathcal{L}(x, \lambda) \leq \inf_{x \in X} \sup_{\lambda \geq 0} \mathcal{L}(x, \lambda)$$

- **Strong duality** restatement:

$$\sup_{\lambda \geq 0} \inf_{x \in X} L(x, \lambda) = \inf_{x \in X} \sup_{\lambda \geq 0} \mathcal{L}(x, \lambda).$$

- Strong duality holds exactly when we can interchange the order of **min** and **max**

Saddle Problem and Game Theoretic Interpretation

Min-Max and Max-Min

Consider the pair of problems:

$$\max_{y \in Y} \min_{x \in X} f(x, y) \qquad \min_{x \in X} \max_{y \in Y} f(x, y)$$

Saddle Problem and Game Theoretic Interpretation

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- Game theoretic interpretation : zero-sum game
- y player maximizes, x player minimizes. Difference is who moves first.

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- For any f, X, Y, the **max-min inequality** (i.e., "weak duality") holds:

$$\max_{y \in Y} \min_{x \in X} f(x, y) \leq \min_{x \in X} \max_{y \in Y} f(x, y)$$

Saddle Problem and Game Theoretic Interpretation

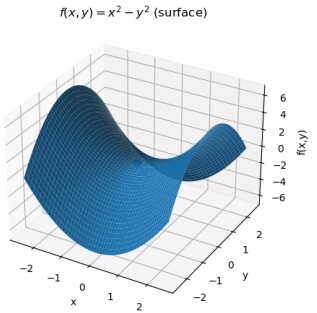
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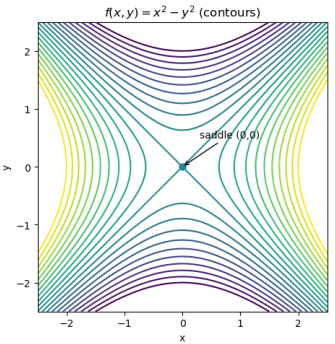
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$$\max_{y \in Y} \min_{x \in X} f(x, y) \qquad \min_{x \in X} \max_{y \in Y} f(x, y)$$

- When do f, X, Y satisfy the **saddle-point property**, i.e., equality holds:

$$\max_{y \in Y} \min_{x \in X} f(x, y) = \min_{x \in X} \max_{y \in Y} f(x, y)?$$





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Sion-Kakutani Theorem

Let $X \subseteq \mathbb{R}^n$ and $Y \subseteq \mathbb{R}^m$ be convex and compact subsets and let $f : X \times Y \to \mathbb{R}$ be a continuous function that is convex in $x \in X$ for any fixed $y \in Y$ and is concave in $y \in Y$ for any fixed $x \in X$. Then,

$$\min_{x \in X} \max_{y \in Y} f(x, y) = \max_{y \in Y} \min_{x \in X} f(x, y).$$

Generalizations possible: Y only needs to be convex (not compact); $f(\cdot, y)$ must be quasi-convex on X and with closed lower level sets (for any $y \in Y$); and $f(x, \cdot)$ must be quasi-concave on Y and with closed upper level sets (for any $x \in X$)

Optimality Conditions

Basic Optimization Problem

We will be concerned with the following optimization problem with variables $x \in \mathbb{R}^n$:

$$
\begin{aligned}
(\mathcal{P}) \ \text{minimize} \quad & f_0(x) \\
& f_i(x) \leq 0, \quad i = 1, \ldots, m \\
& h_j(x) = 0, \quad j = 1, \ldots, s \\
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\end{aligned}
$$

- **Will not assume convexity unless explicitly stated...**

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$$

- **Will not assume convexity unless explicitly stated...**
- **Key Q:** *We have a feasible x. What are the conditions (necessary, sufficient, necessary and sufficient) for x to be "optimal"?*
- First, what can we hope for?
  - **necessary** conditions for the optimality of $x^\star$
  - **sufficient** conditions for the **local optimality** of $x^\star$
- Cannot expect **global optimality** of $x^\star$ without some "global" requirements on f_i, h_j (e.g., convexity)

Complementary Slackness 2.0

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- **If we had strong duality** and $x^\star$ optimal for $(\mathcal{P})$ and $\lambda^\star, \nu^\star$ optimal for $(\mathcal{D})$:

$$f_0(x^\star) = g(\lambda^\star, \nu^\star) = \inf_{x \in X} \Big[f_0(x) + \sum_{i=1}^{m} \lambda_i^\star f_i(x) + \sum_{j=1}^{s} \nu_j^\star h_j(x) \Big]$$

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$$\leq f_0(x^\star) + \sum_{i=1}^{m} \lambda_i^\star f_i(x^\star)$$

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- **Optimality** thus implies **complementary slackness**: $\lambda_i^\star \cdot f_i(x^\star) = 0$, i.e.,

$$f_i(x^\star) < 0 \Rightarrow \lambda_i^\star = 0 \qquad \text{or} \qquad \lambda_i^\star > 0 \Rightarrow f_i(x^\star) = 0$$

Karush-Kuhn-Tucker Optimality Conditions

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- $x^\star \in X$, $\lambda^\star \in \mathbb{R}^m$ and $\nu^\star$ dual variables
- The **Karush-Kuhn-Tucker (KKT) necessary conditions** at $x^\star$ are given by:

Definition: First-Order Necessary KKT Conditions

Karush-Kuhn-Tucker Optimality Conditions

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$$f_i(x^\star) \leq 0,\ i = 1, \ldots, m; \quad h_j(x^\star) = 0,\ j = 1, \ldots, s, \qquad (\text{“Primal Feasibility”})$$

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$$\lambda^\star \geq 0 \qquad (\text{“Dual Feasibility”})$$

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$$f_i(x^\star) \leq 0,\ i = 1, \ldots, m; \quad h_j(x^\star) = 0,\ j = 1, \ldots, s, \qquad (\text{“Primal Feasibility”})$$

$$\lambda^\star \geq 0 \qquad (\text{“Dual Feasibility”})$$

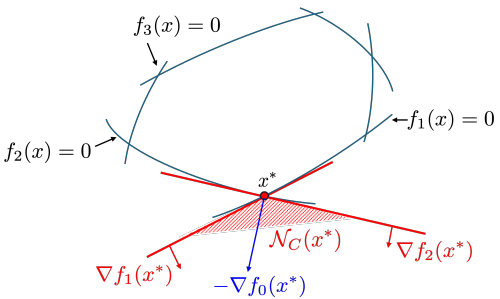
$$\lambda_i^\star f_i(x^\star) = 0, \quad i = 1, \ldots, m \qquad (\text{“Complementary Slackness”}).$$

Geometry Behind KKT Conditions: Inequality Case

KKT Conditions For Case With Inequality Constraints

$$0 = \nabla f_0(x^\star) + \sum_{i=1}^{m} \lambda_i^\star \cdot \nabla f_i(x^\star) \qquad \text{("Stationarity")}$$

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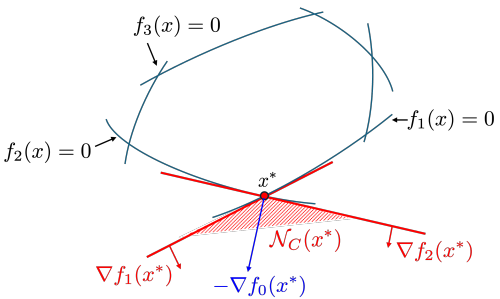


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- Consider all **active** constraints at $x^\star$, i.e., $\{i : f_i(x^\star) = 0\}$
- **Stationarity:** $-\nabla f_0(x^\star)$ is conic combination of gradients $\nabla f_i(x^\star)$ of **active constraints**
- (Complementary slackness: only **active** constraints have $\lambda_i > 0$)
- FYI: $\mathcal{N}_C(x^\star) := \{d \in \mathbb{R}^n : \exists \lambda \geq 0,\ d = \sum_{i=1}^m \lambda_i \nabla f_i(x^\star)\}$ is the **normal cone** at $x^\star$.

Failure of the KKT Conditions

- In some cases, KKT conditions **are not necessary** at optimality

KKT Conditions Failing

$$\min_{x \in \mathbb{R}} x$$
$$x^3 \geq 0.$$

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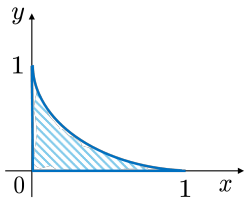
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- **Is this a convex optimization problem? What is $p^\star$? What is $x^\star$?**
- $f_0(x) = x$ and $f_1(x) = -x^3$. Nonconvex because of f_1.
- Feasible set is $[0, +\infty)$; optimal value is $p^\star = 0$, optimal solution $x^\star = 0$.
- KKT condition fails because $\nabla f_0(x^\star) = 1$ while $\nabla f_1(x^\star) = 0$
- There is no $\lambda \geq 0$ such that $-\nabla f_0(x^\star) = \lambda \nabla f_1(x^\star)$.

Failure of KKT Conditions - More Subtle

KKT Conditions Failing

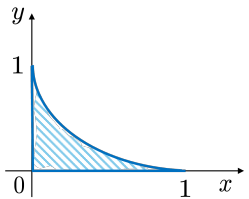
$$\min_{x,y\in\mathbb{R}} \quad -x$$
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- $f_0(x,y) := -x$, $f_1(x,y) := y-(1-x)^3$, $f_2(x,y) := -x$ and $f_3(x,y) := -y$.
- Gradients of objective and binding constraints f_1 and f_3 at $(x^\star, y^\star) := (1,0)$:

$$\nabla f_0(x^\star,y^\star) = \begin{pmatrix} -1 \\ 0 \end{pmatrix}, \quad \nabla f_1(x^\star,y^\star) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \nabla f_3(x^\star,y^\star) = \begin{pmatrix} 0 \\ -1 \end{pmatrix}.$$

- No $\lambda_1, \lambda_3 \ge 0$ satisfy $-\nabla f_0(x^\star,y^\star) = \lambda_1 \nabla f_1(x^\star,y^\star) + \lambda_3 \nabla f_3(x^\star,y^\star)$
- Reason for failing: the linearization of constraint $f_1 \le 0$ around $(1,0)$ is $y \le 0$, which is parallel to the existing constraint $f_3(x,y) := -y \ge 0$

Constraint Qualification Conditions

Setup: $x^\star$ feasible. Active inequality constraints: $I(x^\star) = \{i \in \{1, \ldots, m\} : f_i(x^\star) = 0\}$.

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- all functions $\{f_i : i \in I(x)\}$ in **active** inequality constraints are **convex**
- $\exists \bar{x} \in \operatorname{rel\,int}(X) : f_i(\bar{x}) < 0$ for all $i \in I(x^\star)$

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4. Mangasarian-Fromovitz

- the gradients of equality constraints are linearly independent
- $\exists v \in R^n : v^\mathsf{T} \nabla f_i(x^\star) < 0$ for $i \in I(x^\star)$ and $v^\mathsf{T} \nabla h_j(x^\star) = 0, j = 1, \ldots, s$

Second Order Necessary Conditions

Second Order **Necessary** Optimality Conditions

$x^\star$ feasible for problem $(\mathcal{P})$ and **regular**, $\{f_i\}_{i=1}^m, \{h_j\}_{j=1}^s$ **twice** continuously differentiable in neighborhood of $x^\star$. Let $\mathcal{L}(x; \lambda, \nu)$ denote the Lagrangian function.

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- The Hessian $\nabla_x^2 \mathcal{L}(x^\star; \lambda^\star, \nu^\star)$ of $\mathcal{L}$ in x is **positive semidefinite** on the orthogonal complement $M^\star$ to the set of gradients of active constraints at $x^\star$:

$$d^T \nabla_x^2 \mathcal{L}(x^\star; \lambda^\star, \nu^\star)\, d \geq 0 \text{ for any } d \in M^\star$$

$$\text{where } M^\star := \{d \mid d^T \nabla f_i(x^\star) = 0,\ \forall\, i \in I(x^\star),\ \ d^T \nabla h_j(x^\star) = 0,\ j = 1, \ldots, s\}.$$

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Assume there exist Lagrange multipliers $\lambda^\star \geq 0$ and $\nu^\star$ such that

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$$d^T \nabla_x^2 \mathcal{L}(x^\star;\lambda^\star,\nu^\star) d > 0 \text{ for any } d \in M^{\star\star}$$
$$\text{where } M^{\star\star} := \{d \mid d^T \nabla f_i(x^\star) = 0, \forall\, i \in I(x^\star) : \lambda_i^\star > 0 \text{ and } d^T \nabla h_j(x^\star) = 0, j = 1,\dots,s\}.$$

Then $x^\star$ is locally optimal for $(\mathcal{P})$.

KKT Conditions and Local vs Global Optimality

To summarize...

KKT Conditions and Optimality Notions

- To use the KKT conditions you must first check that one of the constraint qualification conditions holds. Typically, the Slater Conditions might be easiest; the Mangasarian-Fromovitz are the most general from the ones we stated
- If the constraint qualification conditions hold, then:
  - For a **general** optimization problem, the KKT conditions are **necessary** or **sufficient** (depending on which variant you use) for **local optimality** at $x^\star$
  - For a **convex** optimization problem, the KKT conditions are **necessary and sufficient** for **global optimality** at $x^\star$

A Consumer's Constrained Consumption Problem

An Example

Consider a consumer trying to maximize his utility function $u(x)$ by choosing which bundle of goods $x \in \mathbb{R}^+_n$ to purchase. The goods have prices $p > 0$ and the consumer has a budget $B > 0$. The consumer's problem can be stated as:

$$\begin{aligned} &\text{maximize } u(x) \\ &\text{such that } p^{\mathsf{T}}x \leq B \\ &\qquad\qquad x \geq 0, \end{aligned}$$

where $u(x)$ is a concave utility function.

- *Write down the first-order KKT conditions and try to interpret them.*
- *Are these conditions necessary for optimality?*
- *Are these conditions sufficient for optimality?*

A Consumer's Constrained Consumption Problem

$$\begin{aligned} \text{minimize} \quad & -u(x) \\ (\lambda \rightarrow) \quad & p^{\mathsf{T}}x \leq B \\ (\mu \rightarrow) \quad & -x \leq 0, \end{aligned}$$

With $\lambda \in \mathbb{R}_+, \mu \in \mathbb{R}^n_+$ denoting the Lagrange multipliers, the Lagrangian becomes:

$$\mathcal{L}(x, \lambda, \mu) = -u(x) + \lambda(p^{\mathsf{T}}x - B) - x^{\mathsf{T}}\mu.$$

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$\lambda \cdot (p^{\mathsf{T}}x - B) = 0$	("Complementary Slackness" 1)
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Case 1. If the budget constraint is not binding, $p^{\mathsf{T}}x < B$

- $\lambda = 0$ and $\mu_i = 0, \forall\, i : x_i > 0$ (complementary slackness)
- For any $x_i > 0$, we must have: $\frac{\partial u}{\partial x_i} = -\mu_i = 0$
- The consumer purchases the unconstrained optimal amount of each good i

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Case 2.

- $p^{\mathsf{T}} x = B$, then can have $\lambda = 0$ or $\lambda > 0$.
- Case $\lambda > 0$:

$$i : x_i > 0 \quad \Rightarrow \mu_i = 0 \quad \Rightarrow \quad \frac{\partial u}{\partial x_i} = \lambda p_i \quad \Leftrightarrow \quad \frac{\frac{\partial u}{\partial x_i}}{p_i} = \lambda$$

$$i : x_i > 0,\ j : x_j = 0 \quad \Rightarrow \quad \frac{\frac{\partial u}{\partial x_i}}{x_i} = \lambda > \frac{\frac{\partial u}{\partial x_j}}{x_j} = \lambda - \mu_j$$

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- **Bang-for-the-buck** $\frac{\frac{\partial u}{\partial x_i}}{x_i}$ for all **consumed** goods ($x_i > 0$) must be the same, and larger than for **unconsumed** goods

Fenchel Duality

- Elegant and concise theory of optimization duality

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Let $f : \mathbb{R}^n \to \mathbb{R}$. The **conjugate** of f is the function $f^* : \mathbb{R}^n \to \mathbb{R}$ defined as:

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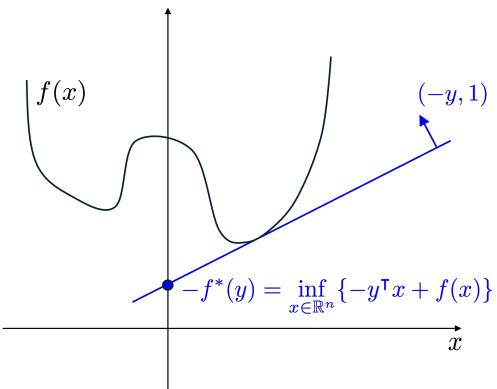
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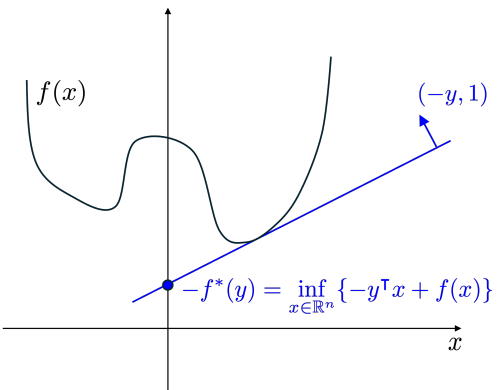
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- **Is f^* convex or concave?**

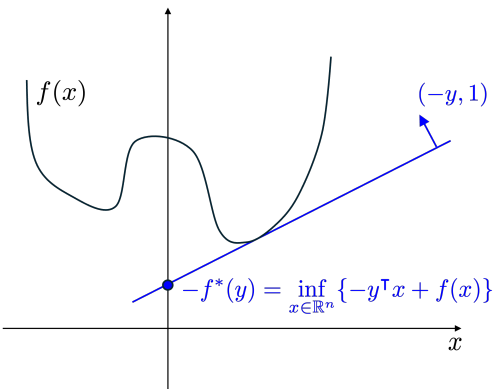
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- If f convex and epi(f) closed, f^* characterizes f in terms of supporting hyperplanes

Conjugates - Examples

$$f^*(y) = \sup_{x \in \mathrm{dom}(f)} \left\{ y^T x - f(x) \right\}$$

The zero function.

For $f(x) = 0$, the conjugate will depend on the relevant domain:

- If $f : \mathbb{R} \to \mathbb{R}$, then
- If $f : \mathbb{R}_+ \to \mathbb{R}$, then
- If $f : [-1, 1] \to \mathbb{R}$, then
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- If $f : \mathbb{R} \to \mathbb{R}$, then $f^* : \{0\} \to \mathbb{R}$ and $f^*(y) = 0$.
- If $f : \mathbb{R}_+ \to \mathbb{R}$, then $f^* : (-\infty, 0] \to \mathbb{R}$ and $f^*(y) = 0$.
- If $f : [-1, 1] \to \mathbb{R}$, then $f^* : \mathbb{R} \to \mathbb{R}$ and $f^*(y) = |y|$.
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$$f^*(y) = \sup_{x \in \mathrm{dom}(f)} \left\{ y^{\mathsf{T}} x - f(x) \right\}$$

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Affine functions.

For $f : \mathbb{R} \to \mathbb{R}$ with $f(x) = a^T x + b$, $f^* : \{a\} \to \mathbb{R}$ and $f^*(a) = -b$.

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What are the conjugates of the following functions?

- $f : (0, \infty), f(x) = -\log x$
- $f : \mathbb{R} \to \mathbb{R}, f(x) = e^x$

Conjugate - Examples

Negative logarithm.

$f : (0, \infty) \to \mathbb{R}$ with $f(x) = -\log x$.

$yx + \log x$ is unbounded above if $y \geq 0$ and reaches its maximum at $x = -1/y$ otherwise. Therefore, $f^* : (-\infty, 0) \to \mathbb{R}$ and $f^*(y) = -\log(-y) - 1$ for $y < 0$.

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Exponential.

$f : \mathbb{R} \to \mathbb{R}, f(x) = e^x$.

$yx - e^x$ is unbounded if $y < 0$. For $y > 0$, $yx - e^x$ reaches its maximum at $x = \log y$, so we have $f^*(y) = y \log y - y$. For $y = 0$,

$$f^*(y) = \sup_x -e^x = 0.$$

In summary, $f^* : \mathbb{R}_+ \to \mathbb{R}$ and

$$f^*(y) = \begin{cases} y \log y - y & y > 0 \\ 0 & y = 0. \end{cases} \tag{1}$$

Fenchel-Young Inequality

Consider the Fenchel conjugate f^* of a function f:

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$$f^*(y) \geq y^{\mathsf{T}} x - f(x)$$

- Having access to f^* allows generating lower bounds on $f(x) \geq y^{\mathsf{T}} x - f^*(y)$

Double Conjugate and Convex Envelope

Consider **the conjugate of the conjugate**, a.k.a. the **double conjugate**, f^{**}:

$$f^{**}(x) = \sup_{y \in \mathbb{R}^n} \{y^T x - f^*(y)\}, \quad x \in \mathbb{R}^n.$$

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Conjugacy Theorem.

Let $f : \mathbb{R}^n \to \mathbb{R}$ be such that epi(f) is closed. Then:

a) $f(x) \geq f^{**}(x)$, *forall* $x \in \mathbb{R}^n$.

b) If f is convex, $f(x) = f^{**}(x)$, $\forall\, x \in \mathbb{R}^n$.

c) $f^{**}(x)$ is the **convex envelope of** f, i.e., epi(f^{**}) is the smallest closed, convex set containing epi(f).

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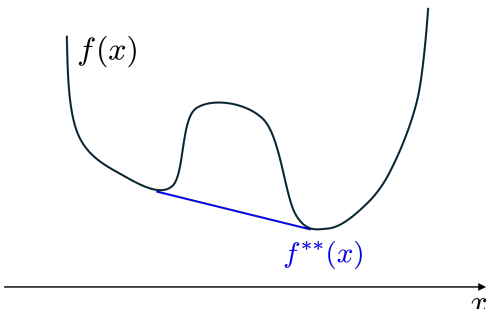
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- The optimal value when minimizing an **arbitrary** f – if finite – equals the optimal value when minimizing the convex envelope of f
- **IF** we had access to f^{**}, we could solve a convex optimization problem to determine the optimal value of any function f
- **Key caveat:** Gaining access to f^{**} is difficult for general f!

Fenchel Duality

Starting Problem.

Consider $f_i : \mathbb{R}^n \to \mathbb{R}$ and $X_i \subseteq \mathbb{R}^n$ for $i = 1, 2$ and the problem:

$$\begin{aligned} &\text{minimize } f_1(x) + f_2(x) \\ &\text{subject to } x \in X_1 \cap X_2 \end{aligned}$$

- Assume optimal value $p^\star$ is finite. Problem can be converted into:

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$$\begin{aligned} g(\lambda) &= \inf_{y \in X_1, z \in X_2} \{f_1(y) + f_2(z) + (z - y)^{\mathsf{T}}\lambda\} \\ &= -\sup_{y \in X_1} \{y^{\mathsf{T}}\lambda - f_1(y)\} - \sup_{z \in X_2} \{-z^{\mathsf{T}}\lambda - f_2(z)\} \\ &= -g_1(\lambda) - g_2(-\lambda), \end{aligned}$$

- **What are $g_1(\lambda)$ and $g_2(\lambda)$ here?**

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- $g_i(\lambda)$ is the conjugate of $f_i(x), i = 1, 2$

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Fenchel Duality

Suppose f_1 and f_2 are convex and **either**
(i) $\operatorname{rel\,int}(\operatorname{dom}(f_1)) \cap \operatorname{rel\,int}(\operatorname{dom}(f_2) \neq \emptyset$
or
(ii) $\operatorname{dom}(f_i)$ is polyhedral and f_i can be extended to $\mathbb{R}$-valued convex functions over $\mathbb{R}^n$ for $i = 1, 2$.
Then, there exists $\lambda^\star \in \mathbb{R}^n$ such that $p^\star = g(\lambda^\star)$ and strong duality holds.